

Banking Loan Analysis Dashboard

A Power BI Analytics Project on Loan Behavior, Credit Risk & Approval Patterns

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Report type **Interactive Power BI Dashboard**

Datasets used **5**

Total records analyzed **295,304**

Report pages **6 (Overview + 5 dataset pages)**

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1. Project Overview

This project brings together five independent loan and credit datasets into a single, connected Power BI dashboard. Rather than analyzing each dataset in isolation, the report is structured as one cohesive product: a landing Overview page summarizing findings across all five datasets, followed by five dedicated pages that each explore one dataset in depth. Clickable navigation buttons and interactive card actions let a viewer move freely between any page, so the report functions like a connected multi-page application rather than five separate reports.

The five datasets do not share common keys or a unified schema — each represents a different loan context (marketing response, credit risk grading, approval screening, legacy applications, and default behavior). The dashboard was therefore built as a multi-fact-table model, with each dataset kept as an independent table and compared visually rather than joined relationally.

1.1 Objectives

- Consolidate five disparate loan/credit datasets into one interactive, navigable dashboard.
- Surface the key rate (default, approval, or take-up) driving each dataset.
- Identify the strongest behavioral or risk drivers within each dataset.
- Present findings through a consistent visual language — a matching color accent, banner, and layout per dataset.

2. Datasets Used

Dataset	Records	Purpose	Accent
Personal Loan Marketing	5,000	Predict which bank customers take up personal loan offers	Blue
Credit Risk	32,576	Assess borrower default risk by credit grade and profile	Green
Loan Approval	2,000	Model approval decisions from applicant financial profile	Orange
Loan Data (Legacy)	381	Classic loan approval demographics dataset	Purple
Loan Default	255,347	Identify default drivers across a large modern loan book	Rose

Total records analyzed across all five datasets: **295,304**.

3. Data Preparation

Each raw file was cleaned before being loaded into the Power BI model:

- Personal Loan Marketing: recoded numeric education levels into readable labels (Undergraduate, Graduate, Advanced/Professional) and labeled binary Yes/No flags.
- Credit Risk: filled missing employment length values with the median, filled missing interest rates using the median within each loan grade, and removed a small number of records with implausible applicant ages.
- Loan Approval: labeled the approval outcome as Approved/Rejected for clearer visuals.

- Loan Data (Legacy): filled missing Gender, Dependents, Self-Employed, Loan Term, and Credit History fields, and labeled the loan outcome.
- Loan Default: labeled the binary default outcome for use across charts and slicers.

Since the five datasets share no common key, no relationships were created between tables in the data model — each remains an independent fact table, compared visually through a shared summary table rather than joined.

4. Dashboard Structure

The report consists of six pages, each reachable from a consistent navigation bar and, on the Overview page, directly clickable through the KPI cards themselves.

4.1 Overview Page

Acts as the landing page and executive summary. Contains five color-coded KPI cards (one per dataset), a single comparison bar chart placing all five rate metrics (take-up, default, approval) side by side, and three headline insight tiles drawn from the underlying data.

4.2 Personal Loan Marketing

KPI cards for total customers and take-up rate; a clustered column chart of education vs. take-up; a bubble/scatter view of income vs. spending; a bar chart of take-up rate by income band; supporting slicers for family size, education, and account type.

4.3 Credit Risk

KPI cards for total applicants, default rate, and average interest rate; a stacked bar of loan intent vs. default status; a column chart confirming the credit grading system (default rate climbs from Grade A to Grade G); a home-ownership comparison; a detail table; slicers for home ownership, grade, and prior default history.

4.4 Loan Approval

KPI cards for total applicants, approval rate, and average credit score; a bubble chart of credit score vs. income; an approval-rate-by-credit-score-band chart; a top-cities table; slicers for city and credit score range.

4.5 Loan Data (Legacy)

KPI cards for total applications and approval rate; property-area and education breakdowns of approval; a detail table of applicant demographics; slicers for gender, marital status, dependents, and property area.

4.6 Loan Default

KPI cards for total loans, default rate, and average DTI ratio; loan-purpose and employment-type breakdowns of default; a bubble chart of credit score vs. interest rate; slicers for loan purpose, employment type, marital status, and co-signer status.

5. Key Findings

Dataset	Headline Rate	Strongest Driver Observed
Personal Loan Marketing	9.6% take-up	Advanced/Professional degree holders take up loans at 13.7% vs. 4.4% for undergraduates — roughly 3x
Credit Risk	21.8% default	Default climbs steadily with credit grade: 10.0% (Grade A) to 98.4% (Grade G)
Loan Approval	44.0% approved	Average applicant credit score of 574 across the dataset
Loan Data (Legacy)	71.1% approved	Highest approval rate of all five datasets
Loan Default	11.6% default	Business loans carry the highest default rate (12.3%); home loans the lowest (10.2%)

The single widest risk spread observed across the entire project is within the Credit Risk dataset: borrowers graded A default at roughly 10%, while borrowers graded G default at over 98% — confirming that the grading methodology is a strong, reliable predictor of repayment behavior.

6. Tools & Methodology

- Data cleaning and preprocessing: Python (pandas) — null handling, categorical recoding, outlier removal.
- Dashboard platform: Microsoft Power BI Desktop.
- Data model: five independent fact tables plus one manually entered summary table used for cross-dataset comparison.
- Navigation: custom button-based page navigation and clickable KPI card actions, replicated across all six pages.
- Visual design: consistent per-dataset accent color, matching banner, canvas background, and KPI card styling used throughout.

7. Conclusion

This project demonstrates how five structurally unrelated datasets can still be unified into one coherent analytical product through consistent visual design and a shared summary layer, rather than a forced data relationship. The resulting dashboard allows a viewer to move from a high-level cross-dataset view down to granular, dataset-specific detail in a few clicks, supporting both quick executive review and deeper analysis within each domain.

8. Author & Project Links

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